Ayush Nadiger

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Education

University of Massachusetts Amherst — Expected May 2027
M.S., Electrical and Computer Engineering — Amherst, MA
Thesis committee: Filip Rozpędek, Don Towsley and Robert Niffenegger.

University of Massachusetts Amherst — May 2026
B.S. with Honors in Computer Engineering — Amherst, MA
B.S. with Honors in Mathematics (Pure Mathematics); B.S. with Honors in Statistics and Data Science.
Minors in Astronomy and Engineering Management. Commonwealth Honors College; Tau Beta Pi.

iCons (Integrated Concentration in STEM). Selected as one of 72 students from across all majors for this interdisciplinary honors certificate built around team research on real-world problems.

Research Experience

Graduate Research Assistant — Sep 2026 – Present
Center for Quantum Networks, UMass Amherst — Amherst, MA

Error correction and routing in two-dimensional hybrid repeater networks

- Extending encoded hybrid repeater chains to 2D networks, where a few hybrid nodes with long-lived memories run error correction and simple repeaters between them generate entanglement, with the aim of choosing node placement, routing, memory and code together. Built stochastic simulations with probabilistic Bell-pair generation, memory decoherence, noisy swaps and multipath reconvergence.
- Developed an exact finite-state description of a region of simple repeaters as seen by its boundary hybrid nodes. It tracks delivery times and correlated phase errors, including those of a partially assembled encoded block, and its size does not grow with the number of rounds.
- Showed that the code can change the best route: on a small network, a detour slightly lowers the delivery objective for unencoded pairs but raises it about 2.2-fold for a three-qubit phase-repetition block.

Memory-assisted GKP-qudit repeaters

- New collaboration with Conall Campbell (UMD) and Matheus Guedes de Andrade on a repeater in which GKP qudits create links between neighboring stations and an outer stabilizer code, run in quantum memories, turns them into fewer, higher-fidelity links. The goal is wider repeater spacing and less squeezing at a given entanglement rate.

Asynchronous Quantum Error Correction — Fall 2026 – Present
Project for CS 648 Quantum Information Science — UMass Amherst

- Studying how to encode Bell pairs into a fixed CSS code as they arrive probabilistically. Each check can be measured only once all of its qubits are present, so the arrival order decides when each error first becomes detectable.
- Proved a lower bound on the error of any decoder, based on the errors that build up before any check can detect them.
- For the $[[72, 12, 6]]$ bivariate bicycle code, showed with Pauli-frame simulation that balancing X- and Z-check completion lowers the twelve-pair block error from about 52% to 20%; decoder-independent lower bounds drop from 4.3–4.8% to 0.08%.

Undergraduate Honors Thesis — 2025 – 2026
UMass Trapped Ion and Photonics Lab; advisors Chris Cox and Robert Niffenegger — Amherst, MA
Geometry-controlled correlated electric-field noise in enclosed ion traps from billiard return spectra.

- Derived how electric-field noise from nearby conducting surfaces is correlated across an array of ions, and how it heats their shared motion.
- For a flat cover above the trap, treated the field as bouncing between the two plates like a billiard ball, and proved that the cover always increases noise perpendicular to the surface and decreases it parallel to it. With the cover at twice the ion height, heating changes by exactly $\zeta(3) \approx 1.20$ and $\eta(3) \approx 0.90$ times the open-trap value.
- Showed that the cover concentrates noise into fewer collective modes: in a ten-ion chain, the strongest mode's share of the noise rises from 23.7% to 28.6%. Connected this to the error of a two-ion entangling gate.
- For curved covers, showed with boundary-element simulations that the shortest reflection path predicts how the noise falls off with distance (correlation 0.9991 across 16 shapes). Also supported the lab's ARTIQ and FPGA-based experimental control.

Summer Undergraduate Research, Mathematics and Statistics REU — Summer 2025
Alternative Models of Mathematical Billiards, with Prof. Chris Cox; funded by a departmental scholarship — UMass Amherst

- Simulated dynamical billiards and compared persistent-homology signatures of chaos with finite-time Lyapunov exponents.
- Proposed three persistent-homology indicators of chaos and mapped them across the two-parameter digon billiard family. They reproduce the Lyapunov-exponent map at about 3–3.7 times lower measured cost and reveal fractal-like boundaries between regular and chaotic regions (box-counting dimension about 1.4).

Other Research Projects

The Bond Repair Problem in Graph States Fall 2025 – 2026
Project for the UMass FTQEC seminar

- Showed that a missing entangling gate and noise on its two endpoint memories look identical to every standard check of the intended graph state, and found exactly which measurements can tell them apart.
- Showed that with the standard checks, the number of copies needed grows as $1/g^2$ as the endpoints' remaining coherence g shrinks, and that for wheel graphs with 8 or more nodes, three simple measurement settings identify any failed edge.
- Confirmed that two settings never suffice by exhaustive search over all 59,049 settings for the 10-node wheel and 218 million setting pairs for the 14-node wheel.

Statistical Memory Tests for Quantum Error-Correction Records Fall 2025 – Winter 2026
Project for STAT 632 Multivariate Statistics

- Developed statistical tests for whether the syndrome record of an error-correcting circuit shows more memory than the circuit itself explains, with guarantees that hold for a finite amount of data. On a simulated correlated-noise example with 10^5 records, the test certifies at least two hidden noise states.
- On 100,000 shots of Google's distance-5 surface-code data, found one detector location that deviates from the standard circuit noise model in both experiments (adjusted $q \approx 0.008$ and 0.011), consistently across time windows and both halves of the data.

Differential Geometry of Quantum Erasure Correction Spring 2025
Project for MATH 563H Differential Geometry, with guidance from Prof. Robert Kusner

- Studied the space of quantum codes geometrically and showed that how a code stops being able to recover from a lost subsystem is controlled entirely by the statistics of that subsystem's state.
- For a lost qubit, showed that correctability breaks down unevenly under small changes, by a factor of exactly $1/C$, where C measures how entangled that qubit is with the rest of the code.

Quantum Annealing for 5G Protocol Testing 2024 – 2025
With Prof. Taqi Raza, UMass ECE

- Formulated 5G protocol-test prioritization as QUBO problems and solved them on D-Wave quantum annealers.

Industry Experience

Quantitative Research Intern Fall 2025
Running Point Capital Advisors El Segundo, CA

- Built path-dependent Monte Carlo models of collateralized fund obligations, including tranche waterfalls, covenants and liquidity gates, to estimate returns, wipeout risk and sensitivities for three deal structures.
- Co-wrote a white paper on the modeling approach and its results.

Talks and Posters

- *Role of Error Correction in 2D Quantum Networks*. Poster, CQN site visit, University of Maryland, October 2026.
- *Geometry-Controlled Noise in Trapped-Ion Systems*. Posters, Massachusetts Undergraduate Research Conference (MassURC) and UMass ECE thesis session, 2026.
- *Differential Geometry of Quantum Erasure Correction*. Talk, mathematics graduate student seminar, UMass Amherst, 2025.

Teaching

Teaching Assistant, UMass Amherst ECE

- ECE 214 Probability and Statistics and ECE 315 Signal Processing Methods, Prof. Dennis Goeckel.
- ECE 201 Analytical Tools for ECE and ECE 213 Continuous-Time Signals and Systems, Prof. Bill Leonard.
- ECE 202 Computational Tools for ECE, Prof. Marco Duarte.

Leadership and Activities

- Vice President, Residence Hall Association, part of the UMass Student Government Association.
- Dry Lab Lead, UMass iGEM (2024 – 2025), leading genetic-circuit modeling; silver medal at the 2025 Grand Jamboree, Paris.

Skills

Python, C/C++, R, MATLAB, FEniCSx, gmsh, Git, Linux, SLURM, LaTeX, ARTIQ.

Relevant Publications and Preprints

[1] **A. Nadiger**. Geometry-controlled correlated electric-field noise in enclosed ion traps from billiard return spectra. arXiv:2608.24770 (2026). Submitted to *Physical Review A*.